Bootstrapping

C8.01-C8.68 CORE ONLY (SKILL IDs)

1 RECOGNISE IT

- Small sample and population distribution unknown/non-normal.
- Need a CI for a statistic without relying on the standard normal/t route.
- A resampling table or index is supplied.

2 DECISION / METHOD

1. Start with the observed sample.
2. Resample n values FROM THAT SAMPLE WITH replacement.
3. Compute the statistic for every resample.
4. Repeat many times and order the statistics.
5. Use the cutoff convention stated/supported by the course or question.

3 FORMULA

`bootstrap CI = lower and upper empirical quantiles of resampled statistics`

4 FORMULA SHEET VS WHAT I KNOW

Gives: no bootstrap formula.
Still know: resampling source, replacement, same n, many repetitions, statistic and tail cutoffs.

5 EXCEL ROUTE

- Go to the named Bootstrap worksheet and identify the original sample/index grid.
- One mean per resample: `=AVERAGE(resample_range)`.
- Order/read the supplied resample means; use stated cutoffs.
- Preserve precision; enter requested limits to four decimals.

6 TINY WORKED EXAMPLE

Observed sample {2,4,7}. One valid n=3 resample is {4,4,2}; its bootstrap mean is 3.333. Repeat to form the bootstrap distribution.

7 TOP EXAM TRAPS

- Resampling from the population/raw database rather than the observed sample.
- Sampling without replacement or changing n.
- Treating one resample as the CI.
- Inventing a universal 250-resample cutoff-index rule.

8 SAY ALOUD

Bootstrap by repeatedly sampling with replacement from the observed sample, then use the empirical statistic distribution.

9 EXIT CHECK

Where do bootstrap draws come from, and are they drawn with replacement?

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1. Named Bootstrap sheet	2. Find original sample	3. Read supplied index/grid
4. Resample WITH replacement	5. One mean per resample	6. Order resample means
7. Use stated cutoff convention	8. No universal 250 rule	9. Enter 4 decimals